

SIVORO

SIVORO MICROFLUIDICS

Living channels. Programmable flow.

The first self-assembling, optogenetically controlled microfluidic platform.

Seed Round · \$1.5M

April 2026

Pre-money \$15M

THE PROBLEM

Microfluidics is stuck in the cleanroom

01 2–4 weeks per chip, \$5k–20k per mask

PDMS fabrication requires a cleanroom. Each design change means a new photomask — slow iteration, high cost, high barrier to entry.

02 Passive channels — no programmability

Active valves require complex external pneumatic systems. No built-in actuation. No design reusability across projects.

03 Single-use waste — clogging kills the chip

Channel fouling is terminal. Each clog means discarding the device and restarting fabrication from scratch.

MARKET OPPORTUNITY

\$25B

Microfluidics market by 2028 — growth limited by fabrication bottlenecks and passive chip designs

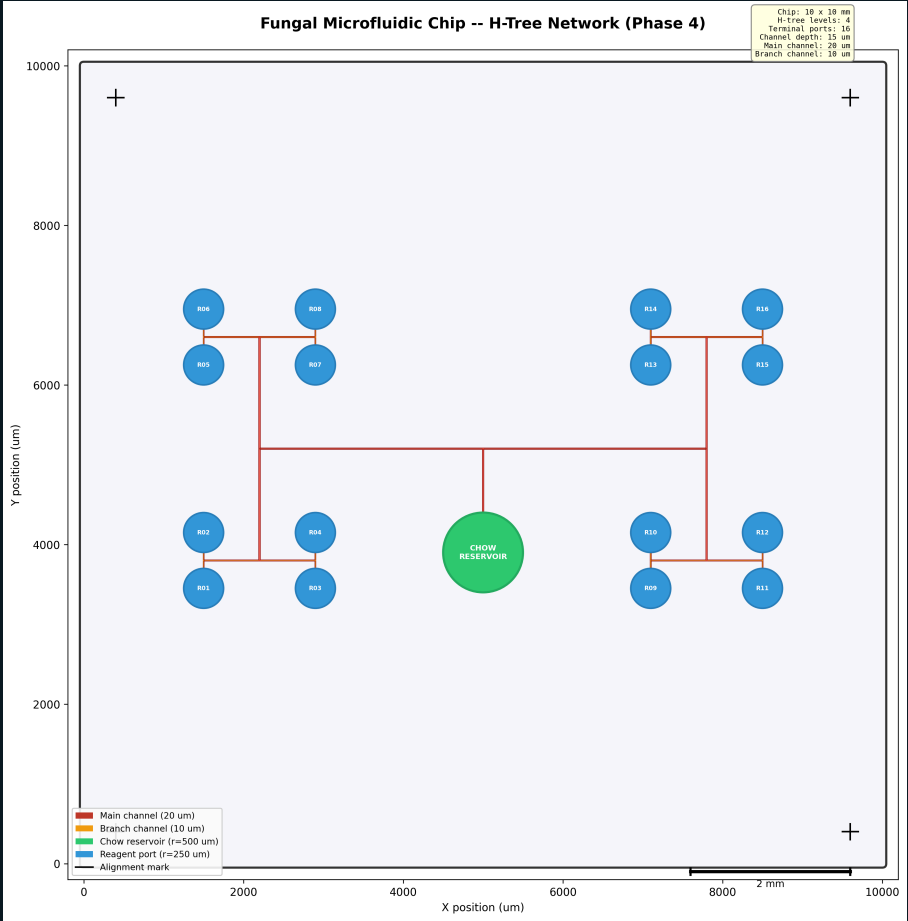
THE CORE BOTTLENECK

Every experiment starts with a cleanroom. Biology moved on — microfluidics hasn't.

THE SOLUTION

One fungus. One week. Zero cleanroom.

FEATURE	SIVORO LIVING CHIP	CONVENTIONAL PDMS
Fabrication	Fungus grows channels	Photolithography (cleanroom)
Valves	Light-controlled (450 nm, built-in)	Pneumatic pumps (external)
Iteration	Days (transform strain)	Weeks (redesign mask)
Channel fouling	Self-clearing (hyphal turnover)	Terminal (chip disposal)
Per-device cost	<\$1 (after strain)	\$50–\$500



H-Tree scaffold — 10×10 mm, 4 levels, 16 ports · channel width 13–20 μm

TECHNOLOGY

Three layers of proprietary IP

LAYER 1 — CHASSIS

Metabolically Orthogonal *N. crassa*

Engineered to eat **xylose only** — cannot metabolize glucose, amino acids, or organic acids present in lab reagents.

Six-gene knockout validated *in silico* from 9,976 combinations screened.

Composition of matter

Screening method

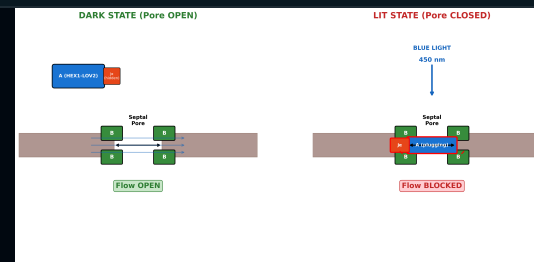
LAYER 2 — VALVE

Optogenetic Septal Pore Valve

Blue light (450 nm) triggers pore closure.

100× dynamic range between dark and lit states.

LOV2-Jα-HEX1 fusion protein + SPA1-ePDZ pore-rim anchor. Two-component system.



LAYER 3 — SCAFFOLD

H-Tree Scaffold & Integration

Parametric design (10×10 mm, 16 ports). Compatible with standard SU-8/PDMS or direct printing.

Channel dimensions engineered for 2–8 hyphae (13–20 μm width, 15 μm depth).

Fungal-specific geometries

Co-culture methods

INTELLECTUAL PROPERTY

Five patent families — provisional ready to file

01	Metabolically orthogonal chassis	Specific genotype + six-gene knockout screening method	HIGH
02	Optogenetic septal pore valve	LOV2-Jα-HEX1 + SPA1-ePDZ constructs & method of flow control	HIGH
03	Living microfluidic device	Chip + organism + scaffold as integrated system	HIGH
04	H-tree scaffold, fungal dimensions	Fungal-specific channel geometry (13–20 μm) novel over PDMS art	MED
05	Six-gene chassis (Δcre-1 + Δwc-1)	Requirement for 450 nm optogenetic activation in orthogonal host	HIGH

FREEDOM TO OPERATE

iLID system (Strickland 2012) — FTO check pending; design-around identified (alternative PDZ pairs).

HEX1 / SPA1 biology — natural *N. crassa* genes, no IP restrictions.

H-tree scaffold — public domain; fungal-specific dimensions are novel.

TIMELINE

File provisional within **30 days** of legal engagement · PCT option at 12 months · Commission FTO search before non-provisional

PRODUCT ROADMAP

From academic kits to platform licensing

TIMELINE	PRODUCT	DESCRIPTION	TARGET MARKET	PRICE POINT
2026-27	Sivoro Start	Strain + scaffold kit · academic labs · drug screening, toxicology	University research labs	\$2,500 / kit
2027-28	Sivoro Organ	Organ-on-chip integration · liver, gut, blood-brain barrier models	Pharma R&D	\$50k-150k platform
2028-29	Sivoro License	IP licensing + custom strains to PDMS chip manufacturers	Microfluidics OEMs	5-10% royalty
2029+	Sivoro Dx	Diagnostic device for point-of-care · RUO then 510(k) pathway	Clinical	Per-assay pricing

TAM EXPANSION

\$500M Academic microfluidics

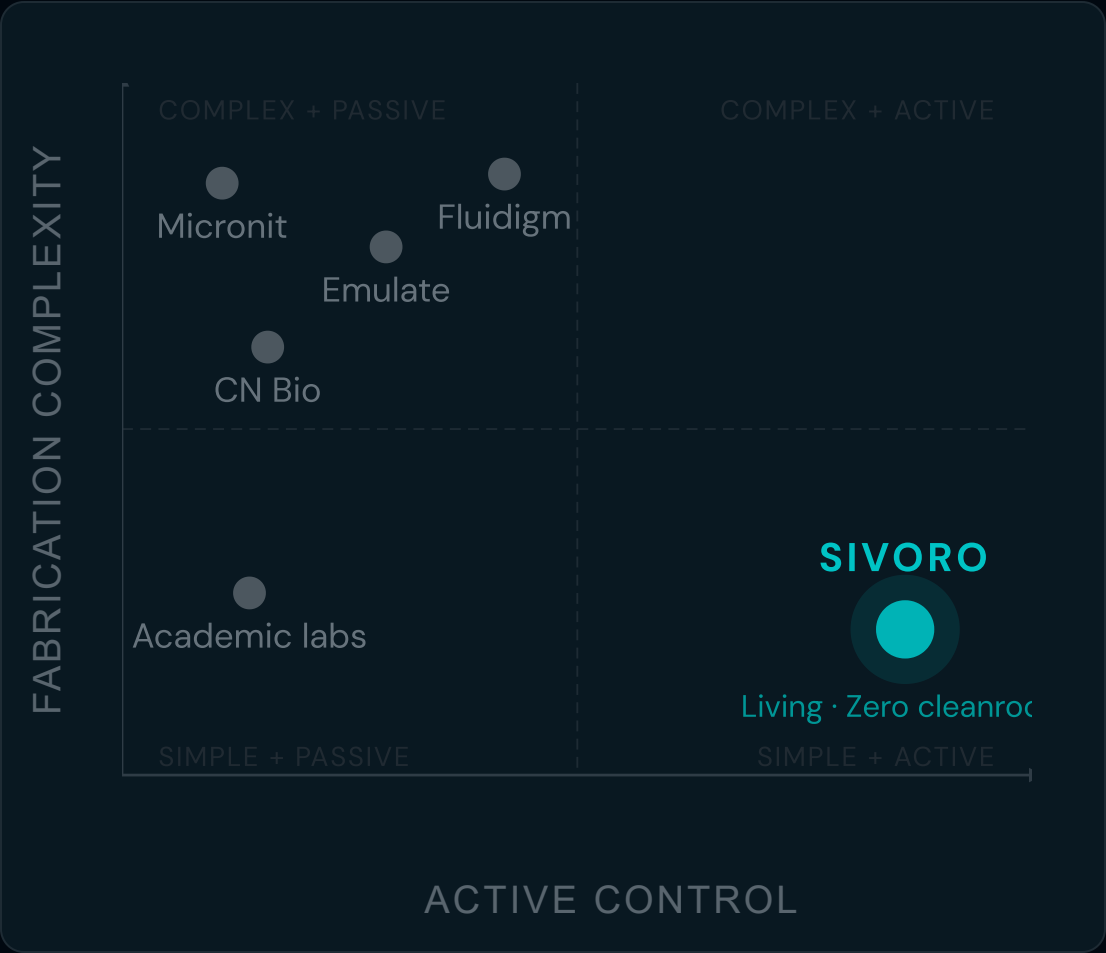
\$2B Organ-on-chip by 2030

\$5B+ PDMS replacement licensing

COMPETITIVE LANDSCAPE

The only **living**, self-assembling microfluidic platform

COMPETITOR	TECHNOLOGY	SIVORO ADVANTAGE
Emulate, Inc.	Organ-on-chip (PDMS + mechanical stretch)	No living channels, no optogenetics; instruments \$200k+
CN Bio Innovations	Organ-on-chip (PDMS)	Sivoro = lower cost, faster design iteration
Micronit	Glass/PDMS fabrication services	Sivoro requires no cleanroom
Academic PDMS labs	Custom fabrication, service model	Sivoro = standard product, not custom service
Fluidigm / Standard BioTools	Integrated fluidic circuits	Price point 10× higher; no biological actuation



BUSINESS MODEL

Three revenue streams, expanding margin over time

STREAM 1

Academic Kits

2026–2028

70%

Sivoro Start kit: strain + scaffold + protocols.

Recurring: media replacements, strain upgrades.

\$2,500 starter · \$500 reagent kit (20 assays)

Gross margin

STREAM 2

Pharma Platforms

2027–2030

60%

Sivoro Organ platform with organ-chip integration.

Custom strain engineering + service contracts.

\$50k–150k platform · \$10k–30k annual service

Gross margin

STREAM 3

IP Licensing

2028+

90%

License strain + valve IP to PDMS manufacturers.

Non-exclusive 5–7% royalty.

Exclusive: 10–12% + upfront.

Pure IP — lowest cost to deliver

Gross margin



DEVELOPMENT STATUS

65% complete — all computational phases done

Metabolic modeling	✓ 100%
KO screen (9,976 pairs)	✓ 100%
Optogenetic valve design	✓ 100%
ColabFold structural validation	✓ 100%
Chip scaffold design	✓ 100%
Wet-lab validation	— 0%
Strain construction	— 0%
SEC-MALS / BLI-SPR	— 0%

12-MONTH MILESTONES (WITH \$1.5M SEED)

Mo.	Milestone	Success Metric
M2	Construct six-gene chassis	PCR-verified knockouts
M4	SEC-MALS on CompA_v3	R_h ≥ 28 nm
M6	BLI/SPR Jα-ePDZ binding confirmed	K_d ≤ 1 μM lit, ≥10× dark/lit ratio
M8	Valve functional (microscopy)	≥50% pore closure at 450 nm
M10	Prototype chip colonization	Fluorescent flow through network
M12	First customer — academic beta site	LOI from 3 labs

TEAM

Founder & key hires

FOUNDER

Michael McKenna

Computational biology · Metabolic modeling · Microfluidics

- Complete computational pipeline — 7,600+ lines Python, genome-scale metabolic modeling, ColabFold structural validation
- Screened 9,976 knockout combinations; designed two validated optogenetic constructs
- Provisional patent draft complete — five claim families ready to file within 30 days

KEY HIRES & ADVISORS NEEDED



Fungal Geneticist

Part-time academic collaboration · *N. crassa* transformation expertise · electroporation + selection



Microfluidics Engineer

First hire post-seed · SU-8, PDMS, organ-chip integration · cleanroom experience



Protein Biochemist

Consultant · SEC-MALS, BLI/SPR execution · *E. coli* / *Pichia* expression

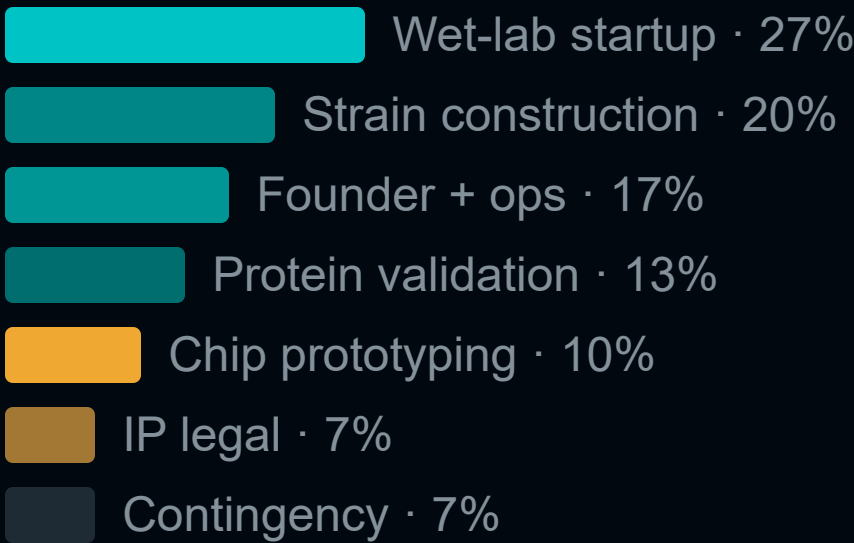
USE OF FUNDS

\$1.5M seed — 18 months to prototype + first sale

CATEGORY	AMOUNT	DESCRIPTION
Wet-lab startup	\$400k	Tissue culture hood, incubators, −80°C, SEC-MALS service contract
Strain construction	\$300k	2 RA salaries · 18 months
Protein expression + validation	\$200k	E. coli / Pichia expression, purification, BLI/SPR
Chip prototyping	\$150k	SU-8 masters, PDMS, photomasks, microscope time
IP legal	\$100k	Provisional + non-provisional + FTO · US + PCT option
Founder salary + ops	\$250k	18 months runway
Contingency (15%)	\$100k	Unforeseen costs

\$1,500k total

ALLOCATION



Next round: Series A · **\$5M** · 2027
Scale production · hire sales · enter pharma pilots

TRACTION TO DATE

Computational validation complete

9,976

Knockout pairs
screened in
silico

2

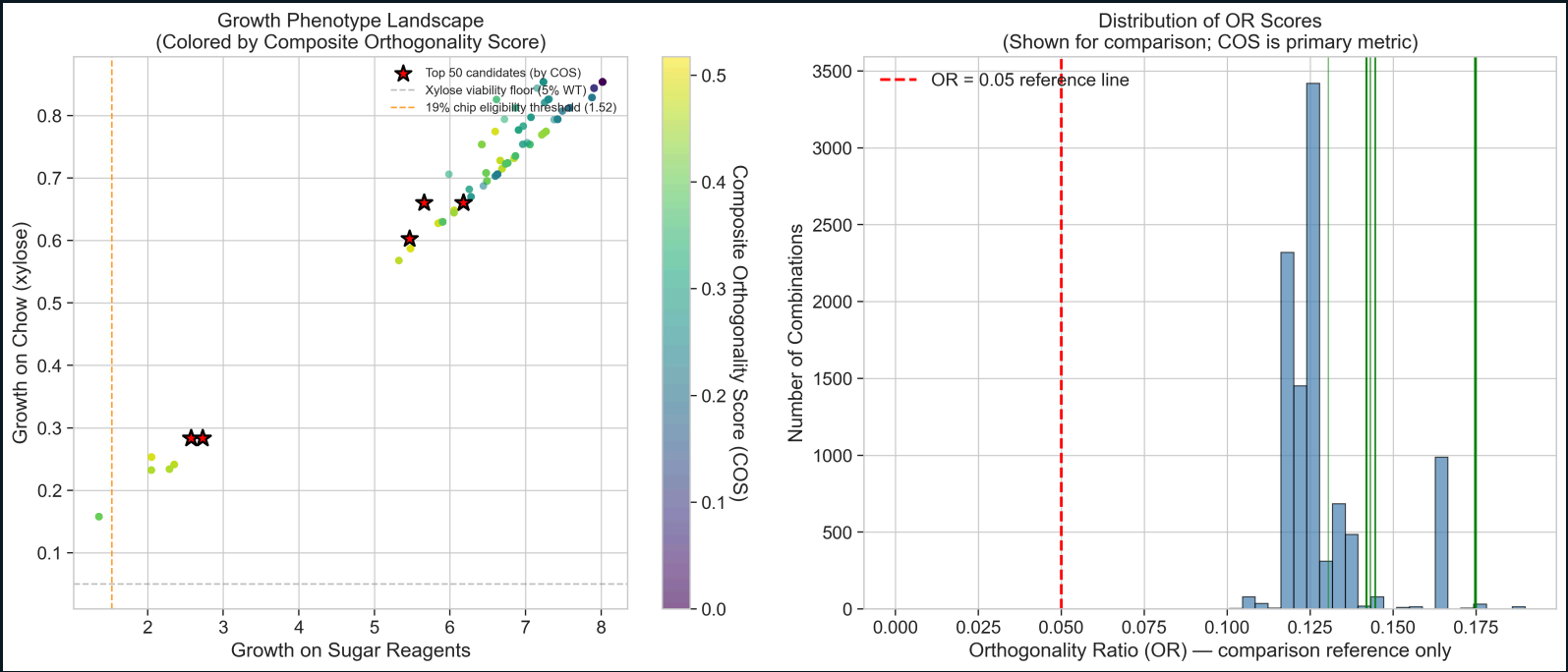
Validated
optogenetic
constructs
(sequence +
structure)

91

pLDDT — LOV2
domain
(ColabFold, high
confidence)

88

pLDDT — HEX1
domain
(ColabFold, high
confidence)



IP STATUS

Provisional draft complete · 5 claim families · ready to file within **30 days** of legal engagement

THE ASK

\$1.5M

Seed Round

Pre-money valuation	\$6M
Post-money valuation	\$7.5M
Runway	18 months
Goal	Prototype + first sale

Key Risks & Mitigations

- HEX1 does not oligomerize
Fallback: full-length HEX1 or engineered self-assembly domain
- Jα C-terminus loses ePDZ binding
Fallback: SpyCatcher/SpyTag — covalent, C-term independent
- Δwc-1 affects growth rate
No published metabolic role; contingency: alternative promoters
- Regulatory delays (FDA for Dx)
Start with RUO products only — academic and pharma R&D, no clearance required

"We are one wet-lab milestone away from a functional prototype — exactly what this seed round funds."